

Junmo Lee

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Research Interests

- I am interested in solving challenging problems in **blockchain systems**. My research covers Layer-1 and Layer-2 scalability, state growth, databases, security, and trusted execution environments (TEE), with a broad interest in exploring new challenges beyond these domains.

Education

PhD	Seoul National University , Electrical and Computer Engineering	Seoul, South Korea
	- Thesis: Optimizing Payment Scalability and Storage Efficiency for Blockchain Systems - Advisor: Prof. Soo-Mook Moon - Virtual Machine & Optimization Lab (VMO)	Mar 2019 – Feb 2026
BS	Seoul National University , Electrical and Computer Engineering	Seoul, South Korea
		Mar 2015 – Feb 2019

Experience

Seoul National University , Postdoctoral Researcher	Seoul, South Korea
Institute of Computer Technology	Mar 2026 – present 3 months

Publications

Ethane: Debloating State Data using Compact Trie for Account-based Blockchain	2026
Junmo Lee, Jaehun Kim, Jiyong Youn, Soo-Mook Moon	
dl.acm.org/doi/10.1145/3767295.3769340 (ACM European Conference on Computer Systems (EuroSys 2026))	
BugSweeper: Function-Level Detection of Smart Contract Vulnerabilities using Graph Neural Networks	2026
Uisang Lee, Changhoon Chung, Junmo Lee, Soo-Mook Moon	
doi.org/10.1609/aaai.v40i1.37021 (AAAI Conference on Artificial Intelligence (AAAI 2026, Oral))	
RouTEE: Secure, Scalable, and Efficient Off-Chain Payments using Trusted Execution Environments	2024
Junmo Lee, Seongjun Kim, Sanghyeon Park, Soo-Mook Moon	
doi.org/10.1109/ACSAC63791.2024.00048 (IEEE Annual Computer Security Applications Conference (ACSAC 2024))	
Trie-Hashimoto: State Trie-Based Proof-of-Work Mining for Optimizing Blockchain Storage	2024
Junmo Lee and Jae-Yun Kim contributed equally.	
Jae-Yun Kim, Junmo Lee, Soo-Mook Moon	
doi.org/10.1109/ACCESS.2024.3360379 (IEEE Access (2024))	
BRAIN: Blockchain-based Inference and Training Platform for Large-Scale Models	2024
Sanghyeon Park, Junmo Lee, Soo-Mook Moon	
doi.org/10.1109/ACCESS.2024.3503573 (IEEE Access (2024))	

Ethanos: Efficient Bootstrapping for Full Nodes on Account-based Blockchain

2021

Jae-Yun Kim, **Junmo Lee**, Yeonjae Koo, Sanghyeon Park, Soo-Mook Moon

dl.acm.org/doi/10.1145/3447786.3456231 (ACM European Conference on Computer Systems (EuroSys 2021))

Awards

ETHGlobal Open Agents Hackathon

May 2026

- [OpenDAIO: Decentralized AI Consensus for High-Stakes Review](#)
- [Finalist \(Top 7 out of 468 projects\)](#)

The Chainlink Fall 2021 Hackathon

Dec 2021

- [Governor-C: Sybil-Resistant Quadratic Voting System](#)
- [Chainlink DAO Prize, Polygon Best DAO/Tooling DApp Prize](#)

ETHCon Korea 2020 Hackathon

Dec 2020

- [Efficient Bootstrapping for Full Nodes on Account-based Blockchain](#)
- [2nd Prize](#)
- [Onther Tokamak Network Contributor Award](#)

Avalanche University Hackathon

June 2020

- [TAVA: Blockchain-based Solution for Micro-mobility Redistribution](#)
- [1st Prize](#)

Energy X Security Hackathon (Ministry of Science and ICT)

Oct 2019

- [Blockchain-based Anonymous Open Micro Grid](#)
- [Excellence Award \(2nd Prize\)](#)